

OPEN SOURCE FOR REGULATED MANUFACTURING: GMP-COMPLIANT IIOT DATA LOGGING

BUILDING SCALABLE, AUDITABLE,
AND COMPLIANT IIOT DATA PLATFORMS WITH
ASF TECHNOLOGIES

Abstract

Pharma and MedTech production require trustworthy industrial data to meet EU Annex 11 and FDA 21 CFR Part 11 compliance. This talk shows how Apache IoTDB and IIoT architectures can serve as modern historians enabling audit trails, ALCOA+ data integrity, electronic records, and secure time-series storage. We present a practical blueprint integrating shop-floor devices with Apache technologies to build scalable, cost-effective, and compliant data infrastructures. Attendees learn how open source empowers regulated manufacturing to modernize operations, unlock insights, and maintain GMP readiness.

Big Pharma Update: Annex 11 & 22 are Here!

- Annex 11 Overhaul: Expanding from 1,500 to 10,000 words, it brings highly detailed requirements for data integrity, cloud computing, and IT security.
- The New Annex 22: Created specifically to regulate AI-supported systems within GMP environments.

Introduction

- Why data integrity matters in regulated manufacturing (Pharma, MedTech, Food)
- Annex 11 & FDA 21 CFR Part 11 in 60 seconds
- The challenge: IIoT data volume vs. compliance demands
- Motivation: Can open source serve as a compliant historian backbone?

Annex 11 & FDA 21 CFR Part 11 in 60 Seconds

us FDA 21 CFR Part 11 (US Regulation)

- Defines rules for electronic records & electronic signatures
- Ensures they are trustworthy, secure, and legally equivalent to paper

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us FDA 21 CFR Part 11 (US Regulation)

- Defines rules for **electronic records & electronic signatures**
- Ensures they are **trustworthy, secure, and legally equivalent to paper**
- Core requirements:
 - Audit trails (time-stamped)
 - User access control
 - System validation
 - Electronic signatures

EU **EU Annex 11 (GMP Guideline)**

- Governs **computerized systems in GMP environments** [\[simplified\]](#)
- Covers the **full system lifecycle**:
 - Validation & risk management
 - Data integrity (ALCOA+)
 - Supplier & system governance
 - Periodic review & monitoring [\[gmpinsiders.com\]](#)

Common Goal

- Ensure **data integrity, traceability, and compliance**
- Enable **audit-ready, reliable digital processes** instead of paper

Key Difference (One-liner)

- **Part 11** = “Are electronic records & signatures trustworthy?”
- **Annex 11** = “Is the entire system validated, controlled, and compliant?”

Takeaway

Both regulations ensure that digital GMP data is trustworthy, —across systems, processes, and the full data lifecycle.

GMP Requirements for IIoT Data

- ALCOA+ principles explained
- Audit Trail, Electronic Records, Electronic Signatures
- Time-series data challenges in GMP (sampling frequency, compression, lineage)
- Why traditional historians struggle or are expensive

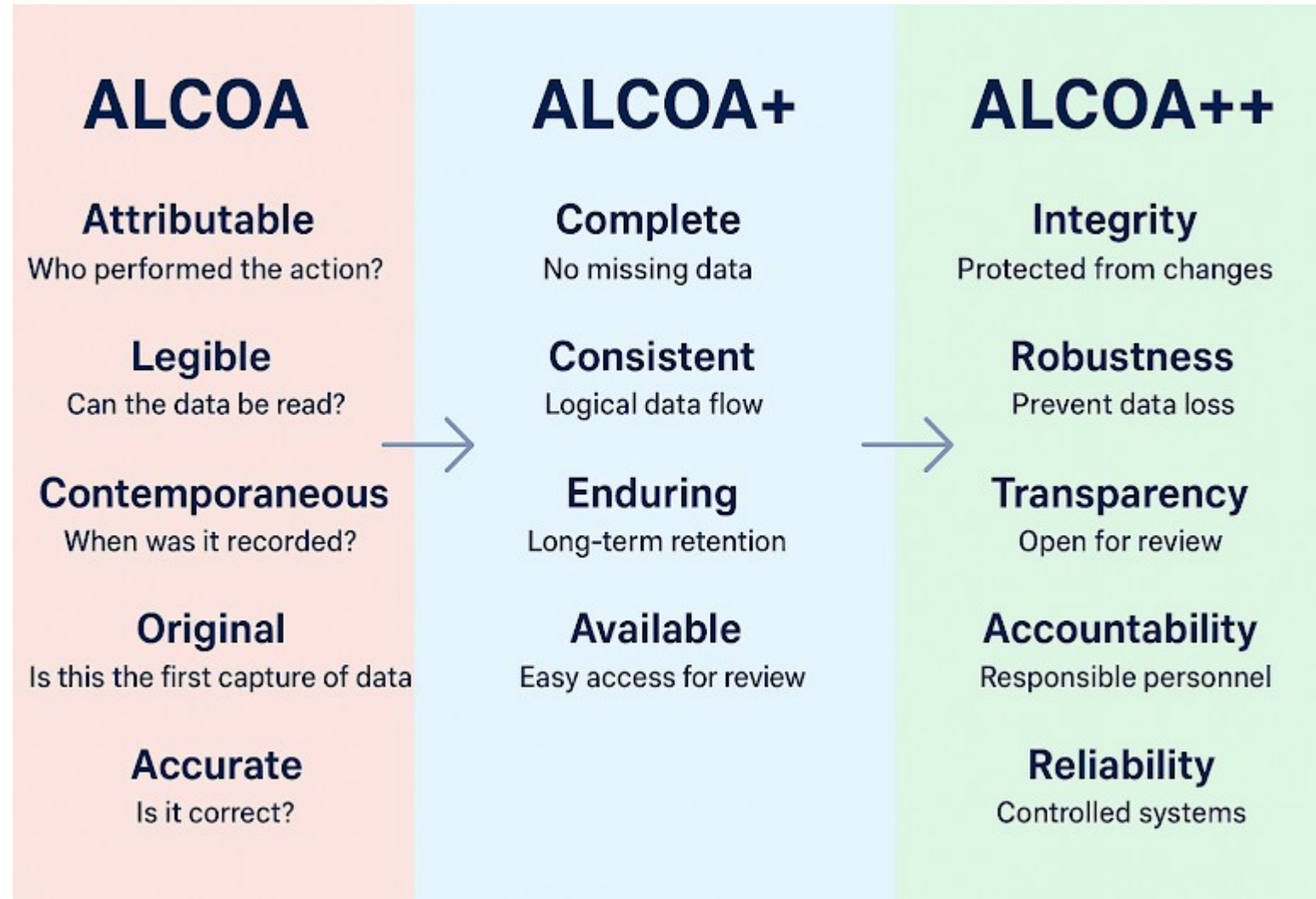
What is ALCOA++ ?

ALCOA+ is a gold standard framework for ensuring data integrity—the accuracy, consistency, and reliability of data—especially in highly regulated industries like pharmaceuticals, biotechnology, and clinical research.

It builds on the original ALCOA principles (Attributable, Legible, Contemporaneous, Original, Accurate) by adding four more: Complete, Consistent, Enduring, and Available.

Together, these principles provide a comprehensive guide for organizations to maintain trustworthy, auditable, and compliant data throughout its lifecycle.





MAIN ELEMENTS OF 21 CFR PART 11



VALIDATION

Systems must be validated to perform as intended



AUDIT TRAILS

All actions must be recorded with date and time stamps



SECURITY

Access to records must be restricted to authorized users



ELECTRONIC SIGNATURES

Signatures must be unique and linked to their respective records



RECORD RETENTION

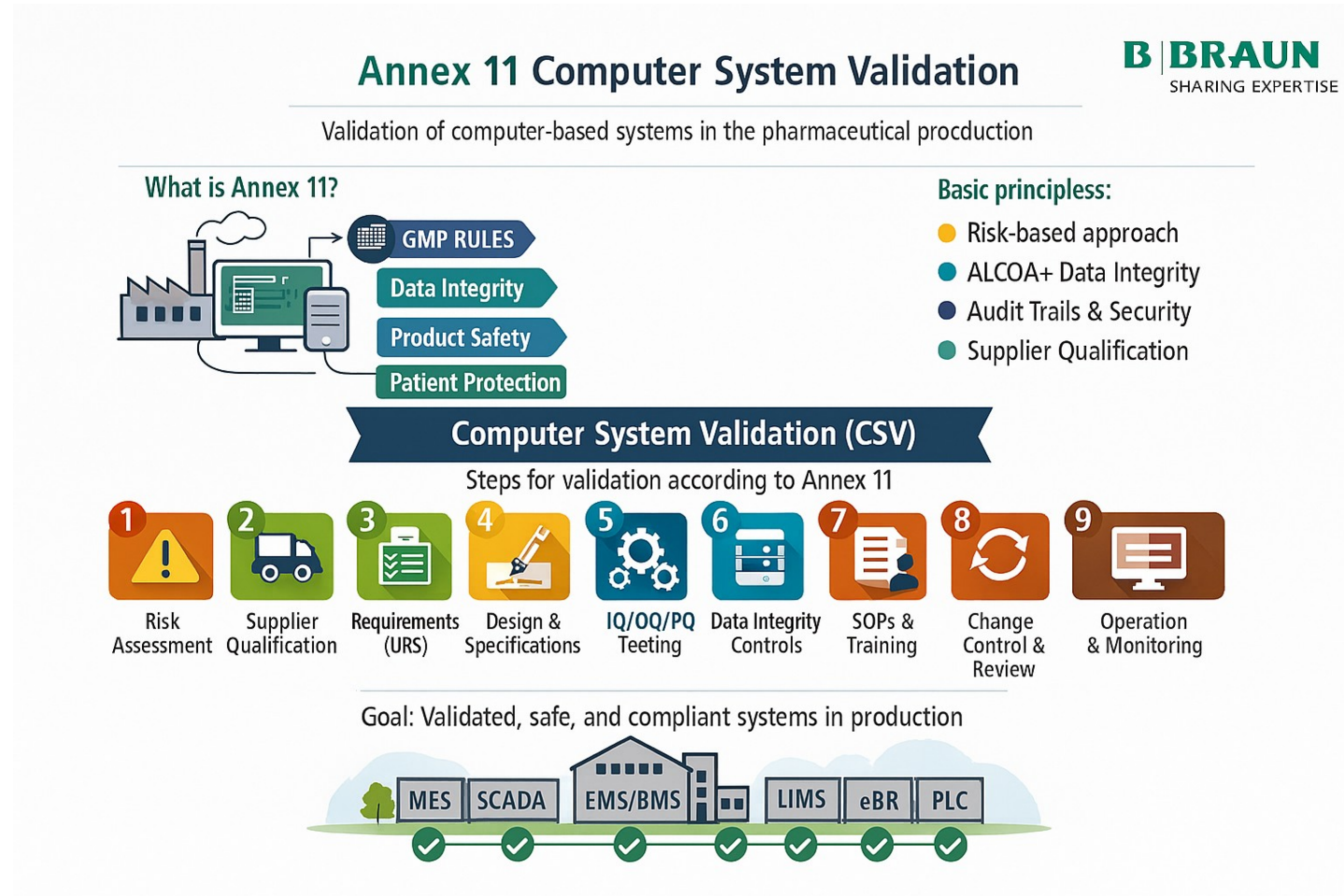
Records must be retained in a secure environment



TRAINING

Personnel must be trained on Part 11 requirements

Annex 11 Computer System Validation



Why Traditional Historians Struggle & Are Expensive

Industrial Data Reality

- High-frequency sensor & PLC data
- Rapid data ingestion & massive volume
- Strict ALCOA+ / audit requirements

Traditional Historian Limitations

- Scaling constraints
→ not built for IIoT data velocity
- Rigid data model
→ limited metadata, lineage, reproducibility
- Weak audit capabilities
→ heavy customization for ALCOA+
- Siloed architecture
→ poor integration with streaming & analytics

Commercial Constraints

- Proprietary licensing (tags / volume-based)
- Vendor lock-in & limited extensibility
- Expensive scaling (storage, HA, retention)
- High validation & customization effort (GAMP 5 context)

The Role of Open Source in Regulated Environments

- Benefits: transparency, extensibility, interoperability, vendor independence
- ASF as a trusted ecosystem
- Community governance = reduced vendor lock-in risk

Open Infrastructure Data Commons (OIDC)

- An open-source software platform for real-time monitoring, data preservation and cross-sector analysis of energy, transport and water infrastructure in low-resource settings.
- [2026 STI Solutions Book \(5\)-compressed.pdf](#)

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Software



N/A



iotdb.apache.org

Apache IoTDB as a GMP-Compliant IIoT Historian

- What IoTDB is (purpose, architecture, performance)
- Data model for industrial sensor & PLC data
- Audit trail capabilities and versioning
- Authentication, authorization, encryption, retention management
- Industrial communication integration (MQTT, OPC-UA via connectors)

Apache IoTDB Selected as One of the Top 60 Global Innovation Cases by the United Nations STI Forum

- <https://www.linkedin.com/pulse/apache-iotdb-selected-one-top-60-global-innovation-cases-o3cdc/?trackingId=FyCa%2FjGICjK2eKBbL7ZHgQ%3D%3D>
- <https://sdgs.un.org/tfm/STIForum2026/Solutionsbook>

Core Enabler: Apache TSFile

- What is TSFile?
- TSFile for Compliance
 - Immutable design patterns for auditability / Data segmentation enabling efficient validation & review
- TSFile + Apache Iceberg
- Hybrid Architecture: IoTDB → TSFile → Iceberg Catalog → Data Lake Analytics
- Advantages for GMP: reproducible queries, tamper-evident storage, lineage

Apache Projects for an End-to-End IIoT Pipeline

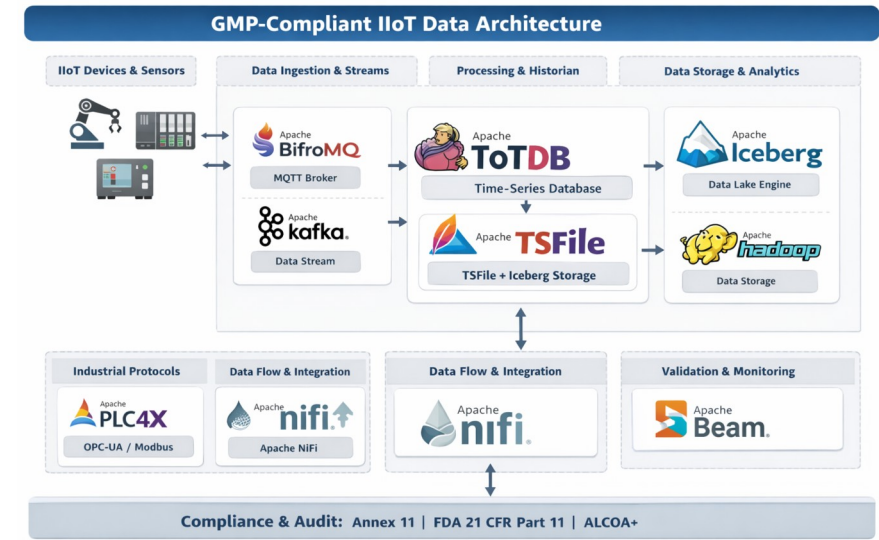
- Messaging & Ingestion
 - Apache Kafka
 - Apache BifroMQ
- Stream Processing & Validation
 - Apache Flink
 - Apache Beam
- Data Integration & Edge
 - Apache NiFi
 - Apache PLC4X
- Data Analytics
 - Apache StreamPipes
 - Apache Superset

Reference Architecture

- A full IIoT + GMP reference architecture combining Apache projects:
- Sensors / PLCs / Edge

→ BifroMQ (MQTT) / PLC4X → Kafka → Flink/Beam processing → IoTDB (hot historian, TSFile) → Iceberg (cold compliant archive) → Superset / BI tools

Include data governance, audit trail storage, retention policies.



Compliance Considerations

- Annex 11 chapter mapping to open-source capabilities
- 21 CFR Part 11 mapping (audit trails, user controls, signature concepts)
- Validation strategy (GAMP 5, system boundaries, IQ/OQ/PQ)

Case Example



Key Takeaways

- Apache ecosystem can support compliant Industrial Data Platforms
- TSFile + Iceberg = powerful future for audit-ready data lakes
- Open source enables scalable, transparent, vendor-neutral GMP architectures

Q & A



THANK YOU!